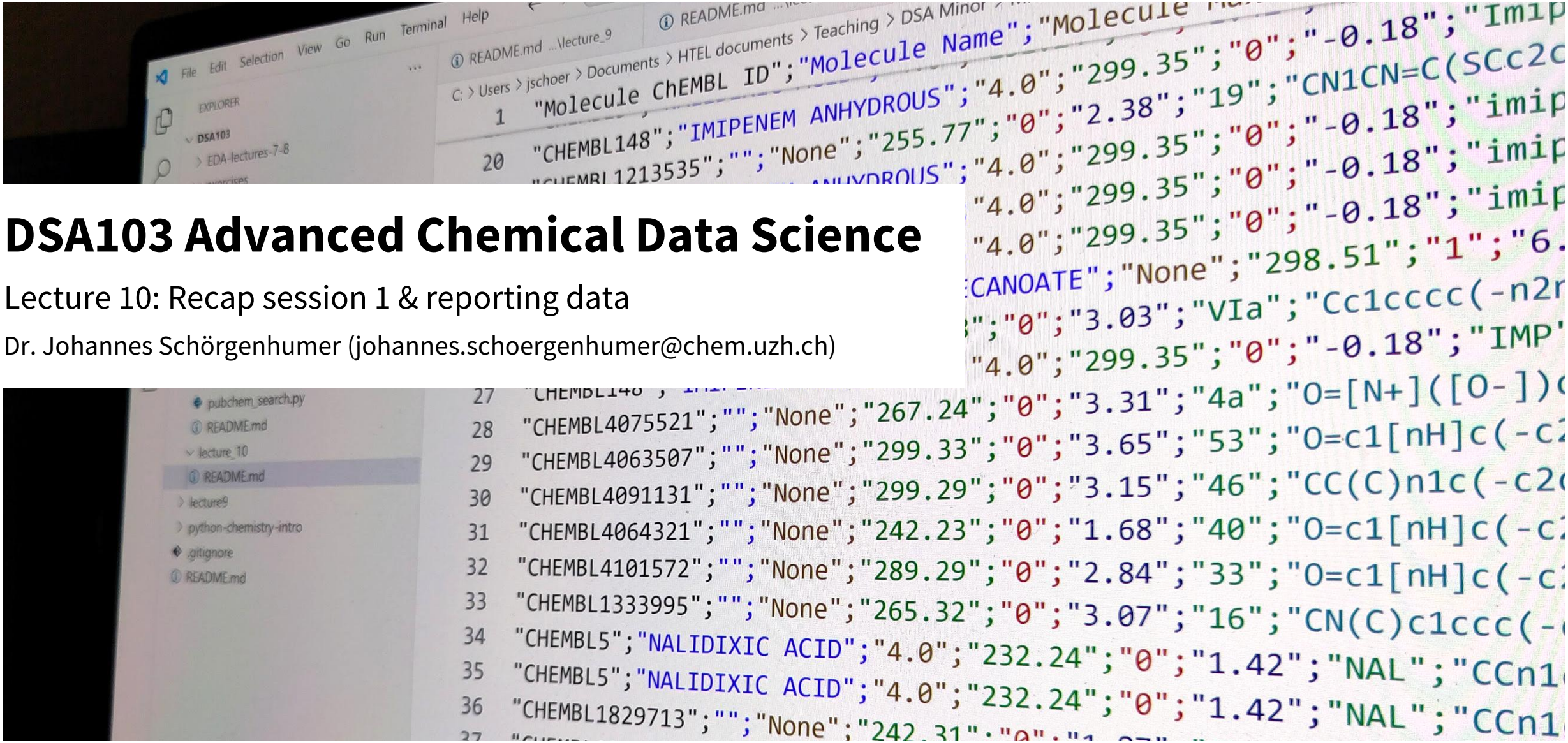
Universität Zürich[UZH]

Department of Chemistry



DSA103 Advanced Chemical Data Science

Lecture 10: Recap session 1 & reporting data

Dr. Johannes Schörgenhumer (johannes.schoergenhumer@chem.uzh.ch)

Content overview

1st part:

— Discussion / solutions assignment_session06

— General feedback Assignments

— Q&A

Info on DSA105

2nd part:

— Light content: Reporting data – FAIR and CARE

General feedback assignments

Collected feedback from first three assignments:

— Errors and exception handling

— Test edge case handling

— Duplicating code vs using functions (or Classes)

— Naming conventions (short vs. verbose)

— Booleans: False vs. None?

— Pass vs. return

— Relative vs. absolute paths

— Code documentation: docstrings vs. single-line / multi-line comments

— Usage of LLMs

Feedback assignments

Please note:

by 7:59 am on the following Friday to Dimitrios by email, with the subject "DSA103_26_Ex1_LastName_FirstName". Your homework will be checked and you will receive feedback; it will not be graded, but you must submit the homework assignments to be admitted to the exam. Please note that one of these homework assignments will be replaced by a small project introduced in session 17 that you have three weeks instead of one week to work on.

Notify us if in case.

This does not imply 100% completing the assignments, but we highly encourage you to do so!

Q&A

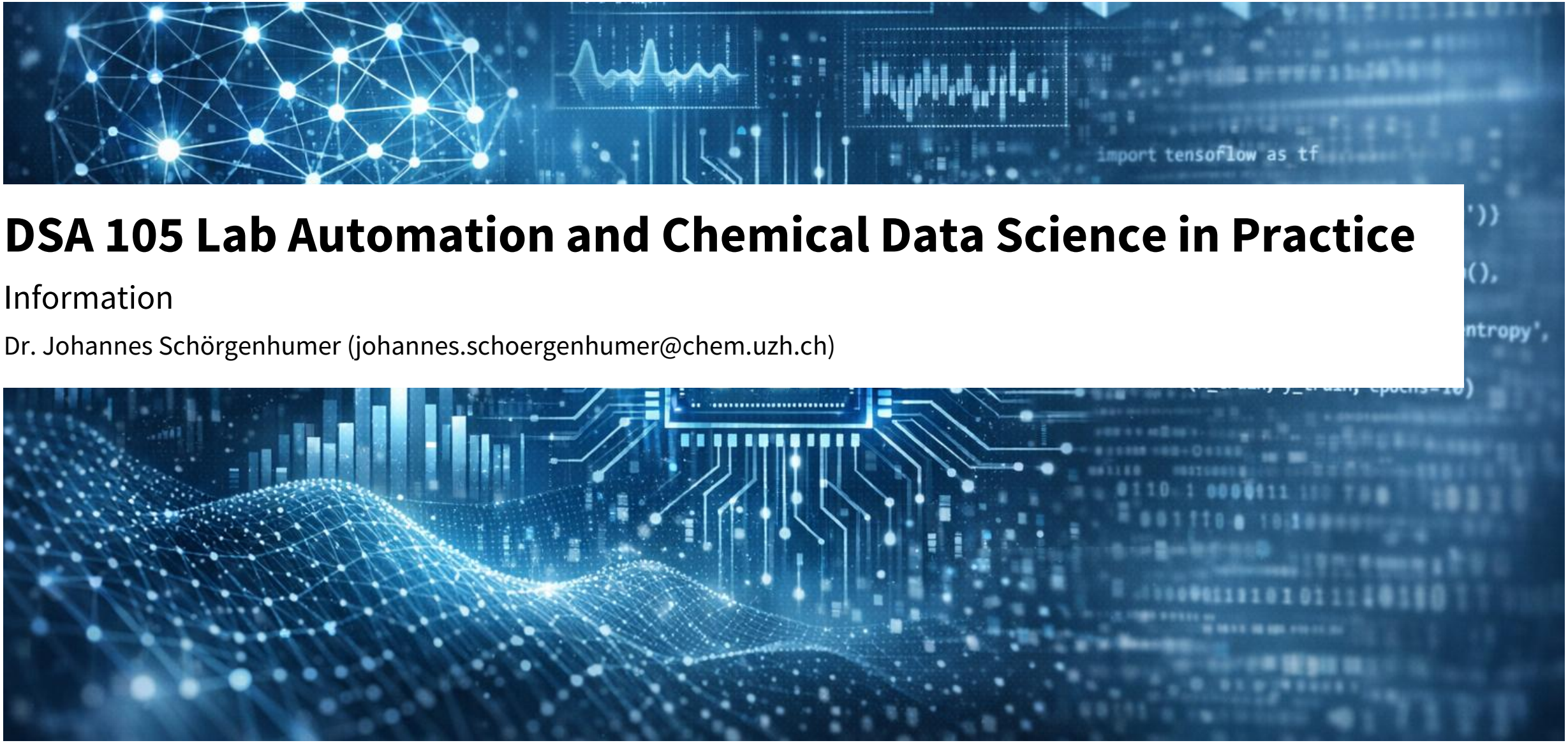
Content?

Organisation?

Feedback?

Universität Zürich[UZH]

Department of Chemistry



DSA 105 Lab Automation and Chemical Data Science in Practice

Information

Dr. Johannes Schörgenhumer (johannes.schoergenhumer@chem.uzh.ch)

DSA 105 Projects

— Projects employ all contents of the DSA minor

— Can be started either in autumn or spring semester

— 2 semesters time to finish the project

— No fixed dates – project work done at home (data part) and upon arrangement in the lab (experiments)

— Done in small groups of 2 or 3 people

How to book the module:

1. Apply to johannes.schoergenhumer@chem.uzh.ch (or in person)
2. State which semester you plan to start
3. And if already known: other group members and topic of interest (can also be done later, once you have seen the data)
4. You will get access to the DSA105 repository
5. If you haven't done so, state your team members and the project you chose

DSA 105 Key idea and steps

Work from home
Lab work upon arrangement

1. You get access to research datasets from UZH – choose one of the projects
2. Process and clean the data, analyse thoroughly, visualise (Content DSA103)
3. Think about possible research questions
4. Project meeting with project supervisor to focus on some ideas to follow up with ML models
5. Use ML models for further data analysis or predictions, etc. (Content DSA104)
6. Project meeting with project supervisor to discuss digital output to validate, test, expand in the lab
7. Lab work to experimentally test hypothesis, collect more data, … (arrange with project supervisors) (Content DSA102)
8. Review of experimental outcome and discuss with project supervisors
9. Hand in deliverables

4 ECTS = 120h of total effort!

DSA 105 Deliverables and grading

Deliverables:

— Well commented python script or Jupyter notebook for the data part (can be referred to in the report)

— Short written report about the story from the dataset to the experimental results (main focus on the discussion of the latter)

Grading:

— Experimental work

— Deliverables

What is expected from these team projects:

— Collaborative effort (fair workload distribution among the team members)

— High degree of self-reliant project planning and execution

— Adhere to good practices on the data part (documentation (markdown, comments, etc), high-quality graphs / plots, use Git/GitHub, …)

— At least one ML model needs to be employed. Choice of models needs to be justified, models should be optimised.

— Deliverables: Scripts/notebooks have to run without errors in a well defined virtual environment (e.g. DSA104 or your own one)

Currently available project topics / datasets

1) Biotransformation dataset (Eawag, supervisor: Jasmin Hafner):

- Dataset (800 entries) on the rate of biotransformation of micropollutants in activated sludge
- Unpublished research data
- **Experimental part only available in Spring Semester!**

2) Macrogarden metabolomics dataset (Schuman group, supervisors: Linus Reichert, Merry Schuman):

- Unpublished research data on chemical diversity in European beech.

3) Multimodal spectroscopy dataset (HTEL, supervisor: Johannes Schörgenhumer):

- Large dataset (790k entries) comprising simulated NMR, IR and MS spectra
- Published dataset (https://zenodo.org/records/14770232)

4) Suzuki coupling dataset (HTEL, supervisor: Johannes Schörgenhumer):

- Small, classic reaction dataset (~300 entries), extracted from an ELN, exclusively manual experiments
- Unpublished research data